

Question	Answer	Marks
5(a)	$\rho = m/V$	C1
	$= (560/9.81)/(1.2 \times 0.018)$	A1
	$= 2600 \text{ kg m}^{-3}$	
5(b)	$(\Delta)p = 940 \times 9.81 \times 1.2$	C1
	(upthrust =) $940 \times 9.81 \times 1.2 \times 0.018 = 200 \text{ N}$	A1
5(c)(i)	tension = $560 - 200$ $= 360 \text{ N}$	A1
5(c)(ii)	$P = Fv$	C1
	$= 360 \times 0.020$	A1
	$= 7.2 \text{ W}$	
5(d)(i)	upthrust decreases	B1
	tension (in wire) increases	M1
	power (output of motor) increases	A1
5(d)(ii)	there is work done (on the cylinder) by the upthrust or GPE of oil decreases (as it fills the space left by cylinder and so total energy is conserved)	B1

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$\Sigma I_{in}(a) = \Sigma I_{out}(a)$	sum of current(a) into junction = sum of current(a) out of junction	B1